Mathematics GCE O Level 2019 - Paper 2

Duration: 2h30 | Cameroon GCE Board

This paper has two sections, A and B. Answer ALL questions in Sections A and B. Section A: answer all 15 questions in the spaces provided; the mark allocation for each question is indicated. Section B: answer all four questions; all questions in Section B carry equal marks (15 marks each). You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, show all the steps in your working, giving your answer at each stage. Calculators are allowed.

Exercise 1 (4 marks)

Evaluate $15 - 12 \div 4 \times 16 + 2$. *[4 marks]*

Exercise 2 (6 marks)

A household consumed 300 units of electricity where the first 220 units are billed at 50 FCFA per unit and the rest at 79 FCFA per unit.

a. Find the number of units to be paid at 79 FCFA.

b. Calculate the total amount paid by this household, in FCFA. *[6 marks for the whole question]*

Exercise 3 (7 marks)

Two cows and two goats cost 560,000 FCFA and one cow and three goats cost 400,000 FCFA.

a. Write two equations in two variables to represent the above statement.

b. Find the cost of each.

c. Find the total cost of a goat and a cow. *[7 marks for the whole question]*

Exercise 4 (4 marks)

Find, to 1 decimal place, the length of an arc PQ which subtends an angle of 60° at the centre of a circle with radius $5.6\ \text{cm}$, taking $\pi = \frac{22}{7}$. *[4 marks]*

Exercise 5 (5 marks)

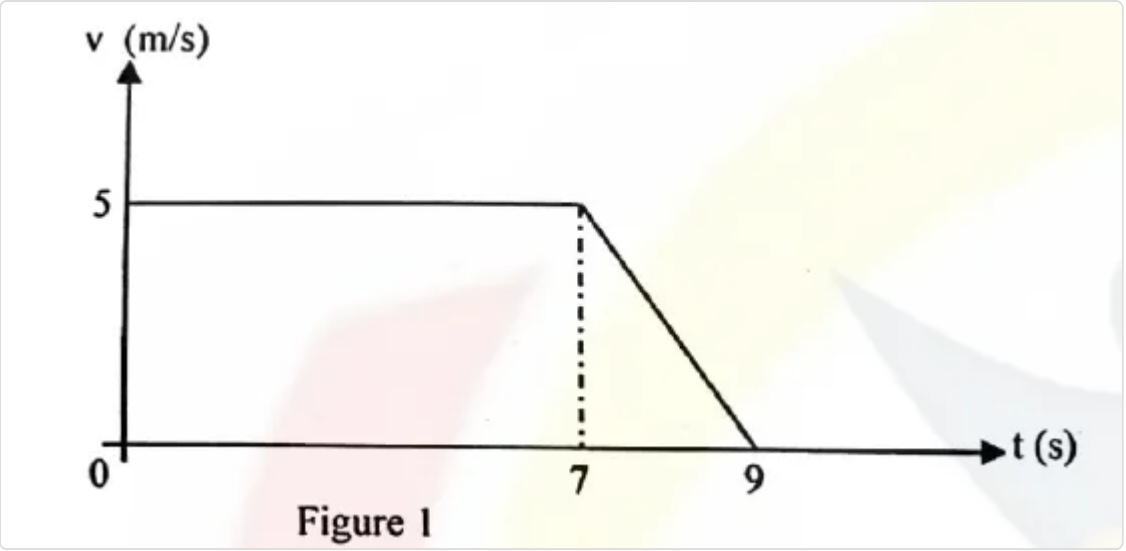


Figure 1

Figure 1 is a $(v\text{-}t)$ graph where the distance covered by a moving object is the area under the graph. Calculate the distance covered in 9 s. *[5 marks]*

Exercise 6 (6 marks)

a. A ship leaves port A on a bearing of 030°. After sailing for $50\,\text{km}$ to Port B, it changes direction on a bearing of 120° and continues in the direction for $40\,\text{km}$ to Port C. Find the distance of C from A, to 2 decimal places.

b. Find the bearing of B from C. *[6 marks for the whole question]*

Exercise 7 (6 marks)

a. Find the range of values of x which satisfy the inequality $x^2 - 5x + 4 \leq 0$.

b. Represent the solution on a number line. *[6 marks for the whole question]*

Exercise 8 (5 marks)

The interior angle of a regular polygon is 108°.

a. Find the value of an exterior angle of this regular polygon.

b. Find the number of sides of this polygon.

c. Write the special name of this regular polygon. *[5 marks for the whole question]*

Exercise 9 (5 marks)

a. Figure 2 shows a road network for towns A, B, C, and M. Determine the number of nodes at M.

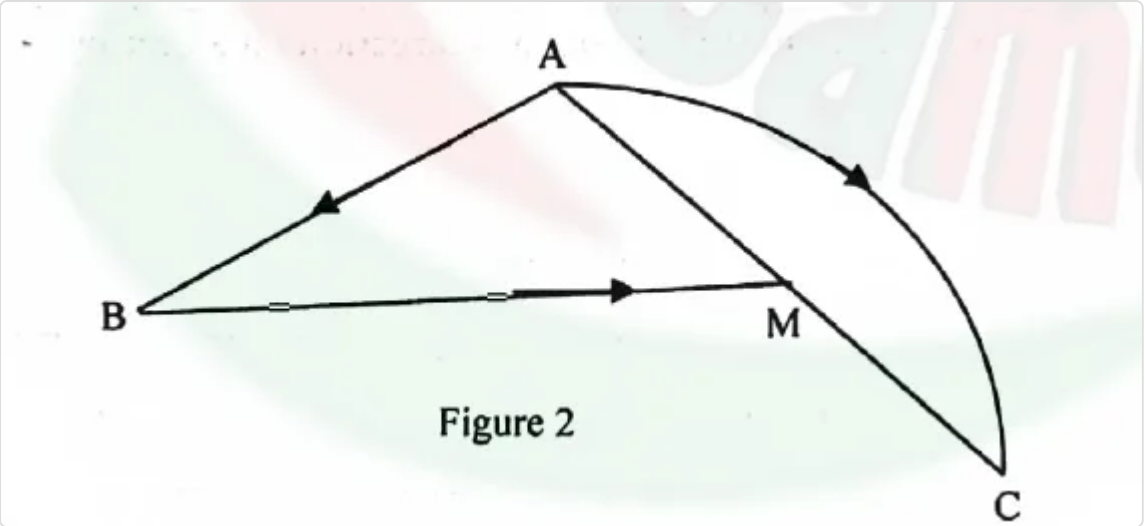


Figure 2

b. Find the number of links between A and B, and between A and C.

c. Find the number of arcs involved in the network. *[5 marks for the whole question]*

Exercise 10 (6 marks)

a. Given that $P = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$, $M = \begin{pmatrix} a & b \\ -1 & 2 \end{pmatrix}$, $N = \begin{pmatrix} -1 & 10 \\ -3 & 10 \end{pmatrix}$ and $PM = N$. Find PM in terms of a and b.

b. Find the values of a and b. *[6 marks for the whole question]*

Exercise 11 (6 marks)

Two statements are such that

p: Peter is studying English Language

q: Peter will live in Britain

a. Translate $p \rightarrow q$ in ordinary English.

b. Write the notation and translate in ordinary English the converse of $p \rightarrow q$.

c. Write the notation and translate in ordinary English the inverse of $p \rightarrow q$. *[6 marks for the whole question]*

Exercise 12 (6 marks)

The sum, S_n, of the first n terms of a sequence of real numbers is given by $S_n = n(n + 5)$.

a. Find the first term.

b. Find the 9th term.

c. Find the sum of the first 10 terms. *[6 marks for the whole question]*

Exercise 13 (6 marks)

Given that $\tan\theta = \sqrt{3}$ and θ is an acute angle.

a. Find $\sin\theta$.

b. Find the value in degrees of θ. *[6 marks for the whole question]*

Exercise 14 (6 marks)

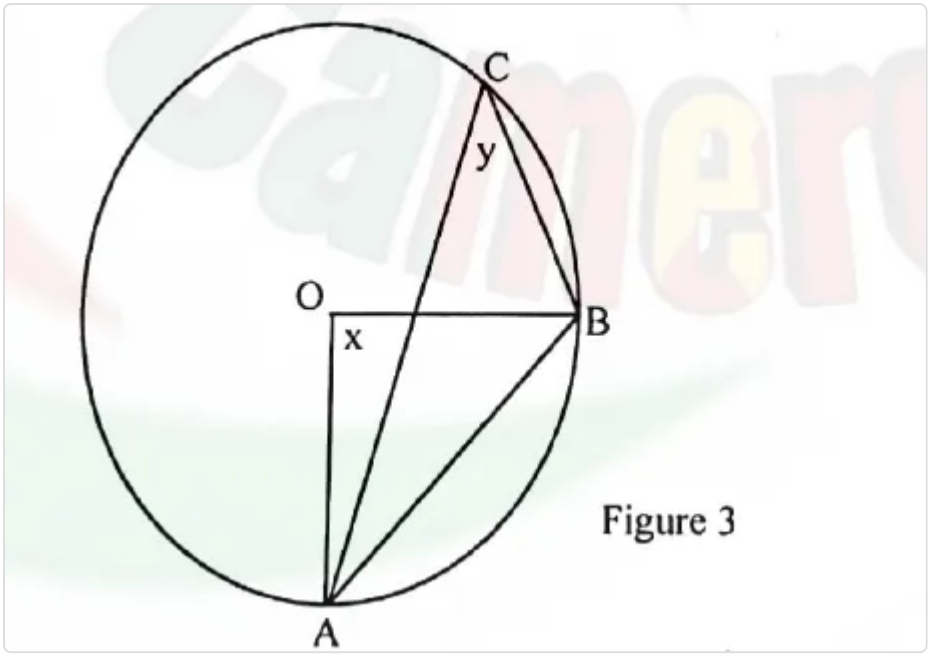


Figure 3 is a circumcircle ABC with centre O and angle $ABO = 50°$. Calculate the angles marked x and y. *[6 marks]*

Exercise 15 (6 marks)

The probability that it will rain on a particular day is $\frac{3}{5}$.

a. Find the probability that it will not rain on that day.

b. Find the probability that it may rain for two consecutive days. *[6 marks for the whole question]*

---

i. Given two functions, $f(x) = x^2 - 4$ and $g(x) = x - 2$. On the same axes and using a scale of $2\ \text{cm}$ to 1 unit on both axes, draw the graphs of $f(x)$ and $g(x)$ for $-3 \leq x \leq +3$. From your graph, determine the values of x at the point(s) of intersection. Write out a quadratic equation whose roots are these values.

ii. A cylindrical tank of radius $3.5\ \text{m}$ and height $10\ \text{m}$ is to be filled with water. Using $\pi = \frac{22}{7}$:

(a) Determine the volume of water when the tank is full.

A tap is attached to the tank such that when opened for 3 minutes, $\frac{1}{5}$ of the water flows from the tank.

(b) Calculate the quantity of water that flows out, in 3 minutes. *[15 marks for the whole question B1]*

---

i. The functions f and g are defined as $f : x \mapsto 3x + 1, x \in \mathbb{R}$ and $g : x \mapsto \dfrac{1}{f(x)}$.

(a) Find $f(2)$.

(b) State the domain of $f(x)$.

(c) Evaluate $g(1)$.

(d) Find $f^{-1}(x)$.

(e) Find $gf(x)$ and state its domain.

ii. Using a ruler and a pair of compasses only:

(a) Construct triangle PQR in which $PQ = 5\,\text{cm}$, $QR = 4\,\text{cm}$ and angle $PQR = 60°$.

(b) Bisect any two sides of triangle PQR.

(c) Construct a circle with centre O, such that the circle passes through the vertices P, Q and R. *[15 marks for the whole question B2]*

---

i. The table below shows the number of pears picked by some children.

No of pears	**1**	**2**	**3**	**4**	**5**
No of children	3	8	4	x	2

(a) Determine the total number of children in terms of x.

Given that the mean is 3,

(b) Find the value of x.

(c) State the modal number of pears.

(d) Determine the total number of children involved.

ii.

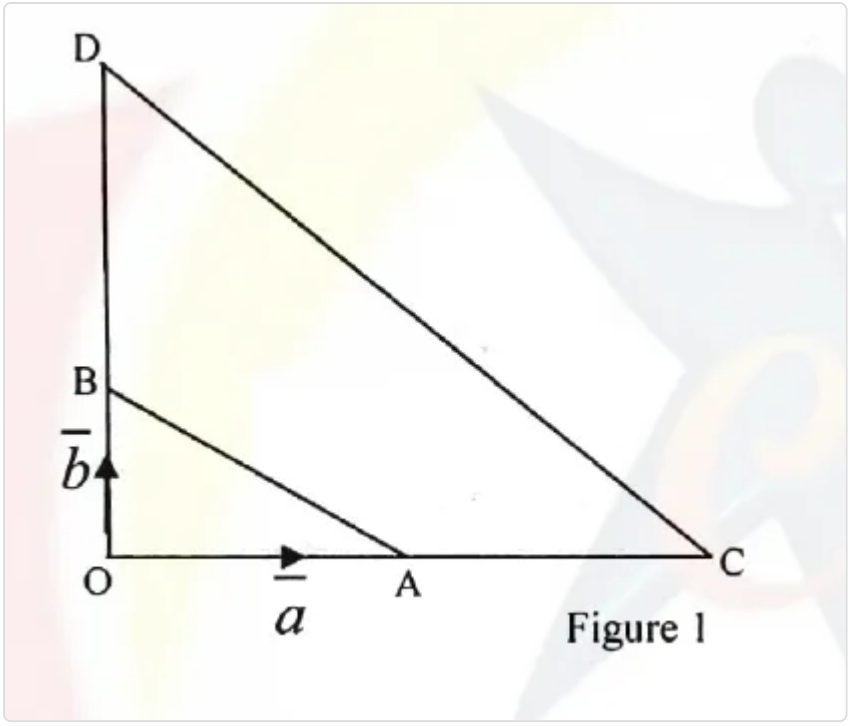


Figure 1

In Figure 1, $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $OD = 4OB$. Find in terms of $\mathbf{a}$ and $\mathbf{b}$:

(a) $\overrightarrow{AB}$

Given that $OA : AC = 1 : 3$, find the vectors

(b) $\overrightarrow{OC}$

(c) $\overrightarrow{CD}$

(d) State the relationship between the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$. *[15 marks for the whole question B3]*

---

i. Bih, Epie and Ngono invested some money in a business and made a total profit of $468\,000$ FCFA. They shared 10% of the profit equally. Find:

(a) The amount shared

(b) The amount each person received

(c) The remainder of the profit

The remainder was shared among Bih, Epie and Ngono in the ratio $3 : 5 : 7$ respectively. Calculate:

(d) Bih's share of the remainder

(e) The total amount received by Bih

ii. In a certain class of a school, the sets M, P and C are defined as:

$M = \{x : x \text{ is a student who studies Mathematics}\}$

$P = \{x : x \text{ is a student who studies Physics}\}$

$C = \{x : x \text{ is a student who studies Chemistry}\}$.

Express the following in ordinary English:

(a) $P \subset M$

(b) $n(M \cap P \cap C) = 25$

Express the following statements in set notation:

(c) 15 students study mathematics and physics but not chemistry

(d) 7 students study Chemistry only.

Given that $n(C) = 37$ and $n(M) = 53$,

(e) Draw a Venn diagram to represent the information above. *[15 marks for the whole question B4]*

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.